Toby Lau

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Education

University of British Columbia — May 2028 (Expected)
B.Sc. Computer Science – GPA: 3.95 — *Vancouver, BC*

Experience

Squarepoint Capital — Montreal, QC
Software Engineer Intern – Portfolio Construction & Trading — *Sep 2026 – Dec 2026*

- Launched a risk analysis platform supporting both Barra and BarraTR models across **8** regions, replacing a **$100k+**/yr vendor.
- Reduced PnL attribution load times by ∼**55%** on a **4.6M**-position book by parallelizing and batching per-date remote calls.
- Sped up per-security factor breakdown from **3–4s** to **<1s** by porting ClickHouse SQL matrix multiplication queries to native q.
- Built kdb+/q endpoints for time-series risk decomposition charts across ∼**200** factors by region, with cross-portfolio deltas.
- Raised per-pod concurrent request throughput **2.2x** across a pool of **16** pods using socket sharding over **3** q processes each.

Software Engineer Intern – Commodities — *May 2026 – Aug 2026*

- Created a monitoring platform for trading desks to schedule, aggregate, and set up alerts on health checks for **1100+** alphas.
- Cut p95 latency **22%** on **35+** prod GraphQL operations by folding a gRPC layer into an Apollo subgraph to drop a network hop.
- Shrank data read per time-series query ∼**72%** across **100+** strategies using adaptive time ranges instead of fixed lookbacks.
- Built a distributed poller/worker data pipeline with Python ARQ to ingest, process, and store ∼**20k** alpha metrics per second.
- Designed a YAML-database sync mechanism enabling **50+** desks to version-control dashboard configurations through GitLab.

Planview — Vancouver, BC
Software Engineer Intern – Planview Copilot — *Jan 2025 – Aug 2025*

- Reduced chat latency **50%+** by batching independent LLM tool calls into parallel AWS Lambda invocations in a Java service.
- Unblocked a major product release for **2k+** users by implementing a dual-token handler during an auth service migration.
- Implemented a log obfuscator/forwarder that routed **21k+** sanitized daily logs to Datadog, retaining raw logs in CloudWatch.
- Delivered a self-service RAG platform for **100+** tenants with S3 and Bedrock, enabling document uploads and analysis/Q&A.
- Prototyped a hybrid tool-calling/MCP LLM chat service and registry to support **8** existing tools while integrating MCP servers.

UBC Department of Computer Science — Vancouver, BC
Undergraduate Teaching Assistant – CPSC 210 (Introduction to Object Oriented Programming) — *Sep 2024 – Dec 2024*

- Taught OOP/OOD, design patterns, and TDD with JUnit to **300+** students; mentored **25+** students in weekly labs/office hours.

APL Zuellig Pharma — Remote
Software Engineer Intern – Distribution Resource Planning — *Oct 2021 – Jan 2022*

- Saved **10+** hrs/week by replacing an Excel stock forecasting workflow with a Python GUI for **46k+** SKUs across **25** warehouses.

Extracurriculars

Databricks — Remote
Student Fellow — *Jun 2026 – Present*

- **1 of 36** selected from **5k+** globally to represent Databricks campus/user groups, with mentorship from Databricks engineers.

Projects

Extreme Vocals Classifier | *Python, PyTorch, NumPy, SciPy, Librosa, Google Colab*

- Synthesized a **500+** extreme vocal sample dataset by isolating vocals with Demucs, extracting and processing relevant vocal regions with NumPy/SciPy (RMS normalization, high-pass filter), and augmenting vocals with librosa from **57+** songs.

Membership Portal | *React, TypeScript, PostgreSQL, Stripe, Supabase, Google Cloud Platform*

- Supported **250+ MAU** and **$5k+** revenue growth by building payments, registration flows, and admin tools from **0**→**1**.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, HTML, CSS, SQL, C, C++, kdb+/q
Frameworks: React, Spring Boot, Next.js, Express, Flask, FastAPI, JUnit, Jest
Technologies: AWS, GCP, Supabase, PostgreSQL, Docker, Kubernetes, ArgoCD, Liquibase/Alembic, GraphQL, gRPC
Tools: Git, Postman/Insomnia, Datadog, Elasticsearch, Jupyter, DBeaver, Vim, Figma, Jira, IntelliJ, Claude Code